PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer

Osijek, Croatia | petra.zlataric@example.com | +385 55 555 0606 | github.com/pzn-ml-example

PROFESSIONAL SUMMARY

Machine Learning Engineer with 5+ years of experience in natural language processing (NLP), document classification, and developing machine learning solutions from research and prototyping through production deployment. Experienced in fine-tuning transformer models, production model serving and monitoring, and MLOps practices using MLflow and Azure. Current doctoral research focuses on processing the Croatian language.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Slavonica Document AI d.o.o. | Osijek *03/2023–Present*

- Developed an automated classification model for incoming documents—including invoices, contracts, and administrative decisions—for public-sector clients, achieving 94% accuracy across 23 classes.
- Fine-tuned BERTić and XLM-R transformer models for named entity recognition in Croatian documents.
- Introduced MLflow for experiment tracking and model registry; deployed models as FastAPI services in Docker containers on Azure ML and Azure App Service.
- Established production model-quality monitoring for data drift, with weekly reporting.
- Mentor two students during their professional internships.

Data Scientist | Baranjska kreditna banka d.d. | Zagreb (remote from Osijek)
09/2021–02/2023

- Developed customer-churn prediction models using gradient boosting and LightGBM for the sales team.
- Analysed customer complaints and classified topics using TF-IDF with logistic regression, later using fastText.
- Created SQL reports using an Oracle data warehouse.

Research Project Assistant | Faculty of Applied Informatics, Drava University
10/2020–08/2021

- Built a corpus of Croatian legal texts and annotated the data.
- Wrote Python scripts to support the research work.

EDUCATION

Doctoral Studies in Computer Science | Drava University *In progress, since 2022*
Research topic: Named entity recognition in Croatian legal texts

MSc in Computer Science (Mag. ing. comp.) | Drava University *2020*

TECHNICAL SKILLS

Python, PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, fastText, spaCy, pandas, NumPy, SQL, MLflow, FastAPI, Docker, Azure ML, Git, Linux

LANGUAGES

Croatian (native) | English (C1) | German (B2)